Vulnerability Assessment Platform

Automated Security Scanning using Python, Nmap & VirusTotal

Project Overview

01

Problem

Manual vulnerability scanning lacks automation and unified threat intelligence

02

Solution

Full-stack platform integrating Nmap port scanning with VirusTotal threat intelligence

03

Outcome

Automated risk scoring engine with real-time alerts and PDF reporting

Problem Statement

Manual Processes

Traditional vulnerability scanning requires manual execution, consuming significant time and resources while introducing human error

Fragmented Tools

Security teams juggle multiple tools for port scanning and threat intelligence without unified risk assessment

No Automation

Absence of automated risk scoring and alerting systems delays critical vulnerability response times

Static Analysis

Limited real-time security assessment capabilities for IP addresses and domains

Proposed Solution

Full-Stack Platform

Complete vulnerability assessment system with React frontend and FastAPI backend

Nmap Integration

Automated port scanning with multiple scan types for comprehensive network analysis

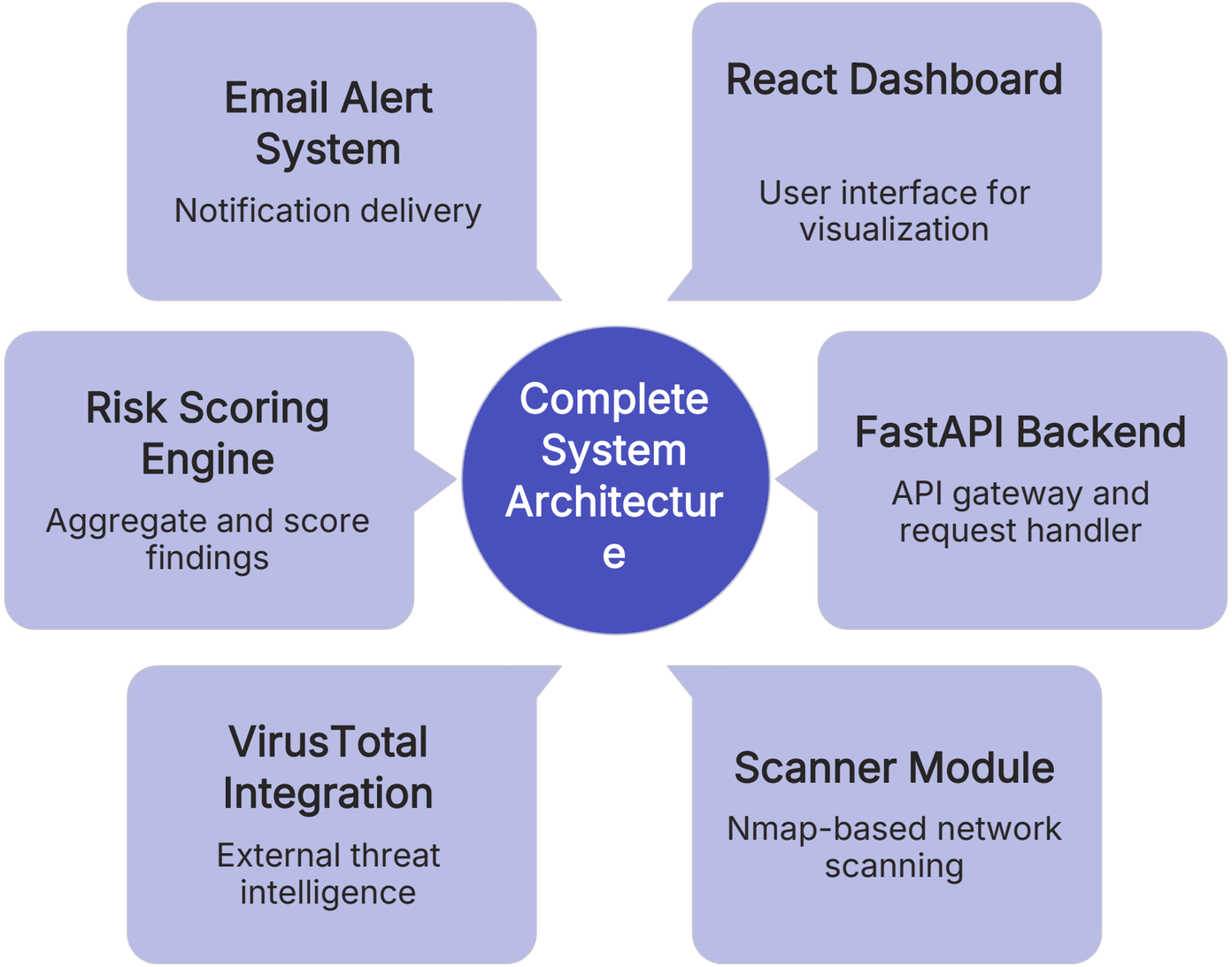
Threat Intelligence

Real-time VirusTotal API integration for malicious detection and reputation scoring

Risk Engine

Custom CVSS-inspired scoring system with automated email alerts and PDF reports

System Architecture



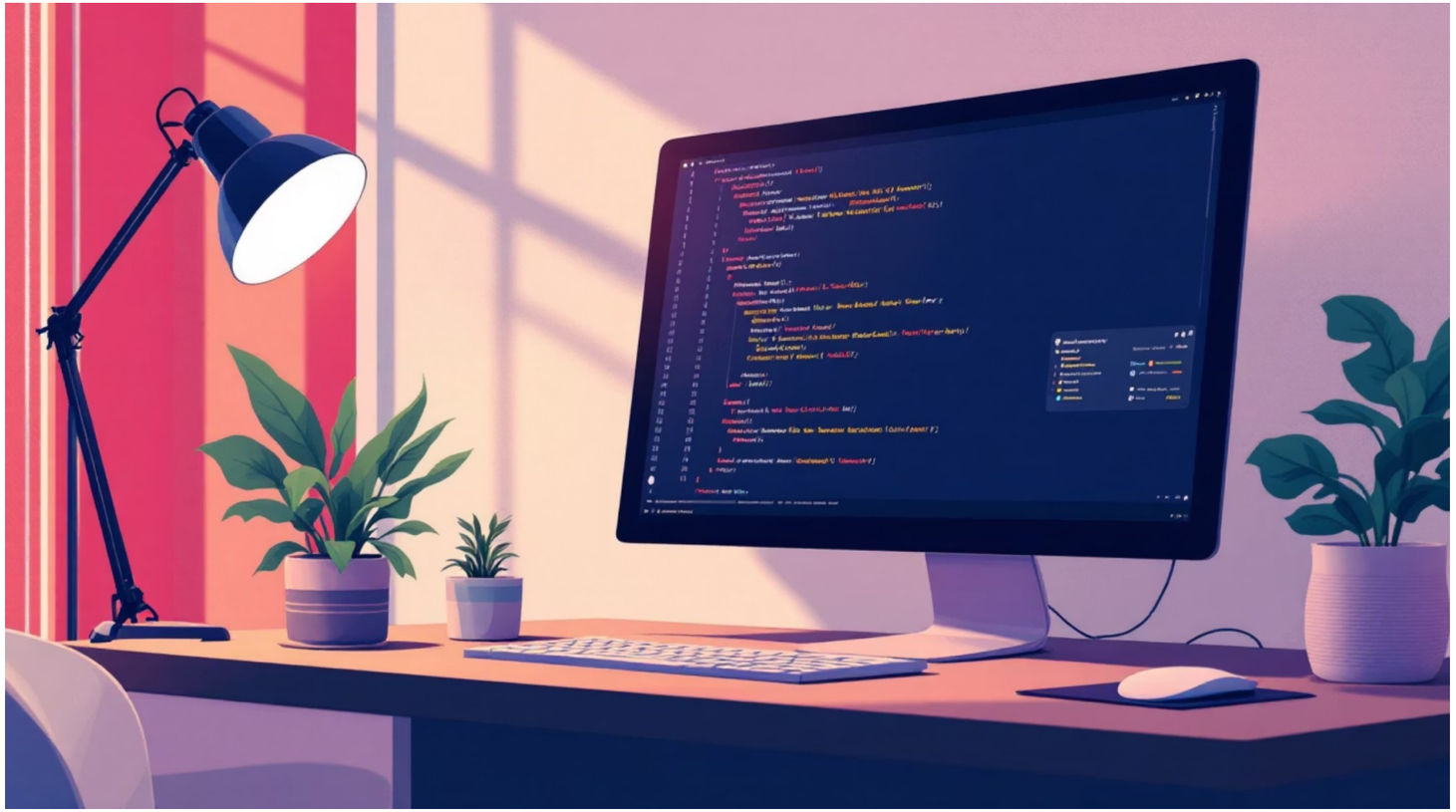
Backend Implementation

FastAPI Core

RESTful API endpoints handle scan initiation, status tracking, and history retrieval with asynchronous task execution for non-blocking operations.

Key Endpoints

- `/api/scan` - Start new vulnerability scan
- `/api/scan/{id}` - Retrieve scan status
- `/api/history` - List past scans



Features

- Pydantic data validation
- Background task processing
- JSON request/response format

Scanning & Threat Intelligence

Nmap Scanner

Port and service detection with multiple scan types: basic, full, and stealth scanning modes

Open Ports Analysis

Identifies open ports, running services, and potential vulnerability vectors in network infrastructure

VirusTotal API

Fetches malicious detection counts, threat reputation scores, and security intelligence from 70+ antivirus engines

Risk Scoring Engine

Input Factors

- Open ports detected
- Dangerous port identification
- VirusTotal malicious counts

Processing

CVSS-inspired weighted algorithm combining multiple security factors into unified risk assessment

Output Results

- Risk score (0-100)
- Severity level
- PDF report generation

Challenges & Learnings

Async Integration

Managing asynchronous scanning operations with FastAPI required careful task coordination and state management patterns

API Error Handling

External API reliability issues with VirusTotal necessitated robust retry logic and fallback mechanisms

Scoring Logic

Designing meaningful risk scoring required extensive research into CVSS standards and security best practices

Real-Time Updates

Implementing live scan progress required WebSocket connections and efficient client-side polling strategies

PDF Generation

Dynamic report creation demanded template design and data formatting that balanced readability with technical detail

Future Enhancements

Persistent Storage

Migrate from in-memory datastore to PostgreSQL for historical data retention and query optimization

Security Layer

Implement JWT authentication and role-based access control for enterprise-grade security compliance

Tool Integration

Add OWASP ZAP for web application scanning and additional security tools for comprehensive assessment

Cloud Deployment

Containerize application with Docker and deploy on AWS/GCP for scalable, distributed scanning capabilities

Thank You